Yan Zhanglu

Research Fellow, National University of Singapore (NUS)

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Summary

Research Fellow at NUS working on **energy-efficient neuromorphic computing**, with a focus on **spiking neural networks (SNNs)**, **hardware-compatible algorithms**, and **hardware-level spike neruon design**. Recent publications span top venues such as ICLR, ICML, IEEE TPAMI, IEEE TNNLS, Acs Synth.Bio, WWW, ASPLOS, and AAAI.

Research Interests

Neuromorphic computing; spiking neural networks; quantization and mixed-precision inference; transformer/LLM-efficient inference; algorithm–hardware co-design.

Work Experience

National University of Singapore (NUS), Singapore — *Oct 2024 – Present*

Research Fellow, mentored by Prof. Weng-Fai Wong

- Neuromorphic algorithms and architecture-aware learning/inference for energy-efficient AI.
- LLM for biomedical and biological applications.

Education

National University of Singapore (NUS) — *Ph.D., Computer Science, 2024*
National University of Singapore (NUS) — *M.Sc., Artificial Intelligence, 2020*
Xi'an Jiaotong University (XJTU) — *B.Sc., Computer Science, 2019*

Selected Publications

1. ***Zhanglu Yan***, Jiayi Mao, Qianhui Liu, Fanfan Li, Tao Luo, Gang Pan, Bowen Zhu, Weng-Fai Wong. Otters: An Energy-Efficient Spiking Transformer via Optical Time-to-First-Spike Encoding. *(ICLR 2026, CAAI-A)*. ***Corresponding author, co-first author***.

2. ***Zhanglu Yan***, Shida Wang, Kaiwen Tang, Zhenyu Bai, Weng-Fai Wong. HyperSNN: A new efficient and robust deep learning model for resource constrained control applications. *(ACC 2026, Oral, Tsinghua A, CAA-A)*. ***First author***.

3. Chenyu Wang, ***Zhanglu Yan***, Zhi Zhou, Xu Chen, Weng-Fai Wong. Energy-Efficient and Dequantization-Free Q-LLMs: A Spiking Neural Network Approach to Salient Value Mitigation. *(WWW 2026, CCF-A)*. ***Co-corresponding author***.

4. Zhenyu Bai, Pranav Dangi, Rohan Juneja, Zhaoying Li, ***Zhanglu Yan***, Huiying Lan, Tulika Mitra. A Data-Driven Dynamic Execution Orchestration Architecture. *(ASPLOS 2026, CCF-A)*.

5. Kaiwen Tang, ***Zhanglu Yan***, Weng-Fai Wong. Sorbet: A Neuromorphic Hardware-Compatible Transformer-Based Spiking Language Model. *(ICML 2025, CCF-A)*. ***Corresponding author***.

6. ***Zhanglu Yan***, Kaiwen Tang, Jun Zhou, Weng-Fai Wong. Low Latency Conversion of Artificial Neural Network Models to Rate-encoded Spiking Neural Networks. *(TNNLS 2025, CAAI-A, JCR-Q1)*. ***First author***.

7. Hanyu Yang, ***Zhanglu Yan***, Huacheng Zhu, Bowen Yan, Wei Chen, Daming Fan. Numerical Solution-Based Algorithm Assisted Development of a Continuous Flow Microwave Reactor. *(Chemical Engineering Journal 2024, JCR-Q1)*.

8. Kaiwen Tang, ***Zhanglu Yan***, Weng-Fai Wong. OneSpike: Ultra-low latency spiking neural networks. *(IJCNN 2024, CORE-B)*. ***Corresponding author, co-first author***.

9. ***Zhanglu Yan***, Weiran Chu, Yuhua Sheng, Kaiwen Tang, Shida Wang, Yanfeng Liu, Weng-Fai Wong. Integrating Deep Learning and Synthetic Biology: A Co-Design Approach for Enhancing Gene Expression

via N-Terminal Coding Sequences. *(ACS Synth. Biol. 2024, JCR-Q1)*. ***Co-corresponding author, co-first author***.

10. ***Zhanglu Yan***, Jun Zhou, Weng-Fai Wong. CQ+ Training: Minimizing Accuracy Loss in Conversion from CNNs to SNNs. *(IEEE TPAMI 2023, CCF-A, JCR-Q1)*. ***Corresponding author, Co-first author***.
11. ***Zhanglu Yan***, Shida Wang, Kaiwen Tang, Weng-Fai Wong. Efficient Hyperdimensional Computing. *(ECML 2023, Oral, CORE A)*. ***First author***.
12. ***Zhanglu Yan***, Jun Zhou, Weng-Fai Wong. EEG Classification with Spiking Neural Network: Smaller, Better, More Energy Efficient. (*Smart Health 2022*). ***First author***.
13. ***Zhanglu Yan***, Jun Zhou, Weng-Fai Wong. Near lossless transfer learning for spiking neural networks. *(AAAI 2021, CCF-A)*. ***Corresponding author, Co-first author***.
14. ***Zhanglu Yan***, Jun Zhou, Weng-Fai Wong. Energy Efficient ECG Classification with Spiking Neural Network. *(Biomedical Signal Processing and Control 2021, JCR-Q1)*. ***First author***. High citation (over 150).

Recent Work and Focus

(On submission / preprint.)

1. ***Zhanglu Yan***, Kaiwen Tang, Zixuan Zhu, Zhenyu Bai, Qianhui Liu, Weng-Fai Wong. *Matterhorn: Efficient Analog Sparse Spiking Transformer Architecture with Masked Time-To-First-Spike Encoding*. arXiv:2601.22876, 2026. ***First author***.
2. Chenyu Wang, ***Zhanglu Yan***, Zhi Zhou, Xu Chen, Weng-Fai Wong. *Kirin: Improving ANN Efficiency with SNN Hybridization*. arXiv:2602.08817, 2026. ***Co-corresponding author***.
3. Kaiwen Tang, Jiaqi Zheng, Yuze Jin, Yupeng Qiu, Guangda Sun, ***Zhanglu Yan***, Weng-Fai Wong. *SpikySpace: A Spiking State Space Model for Energy-Efficient Time Series Forecasting*. arXiv:2601.02411, 2026. ***Corresponding author***.
4. Fanfan Li, ***Zhanglu Yan***, Jiayi Mao, Guolei Liu, Huihui Ren, Bangbang Qin, Zhongfang Zhang, Haiyue Zhang, Yiyang Shen, Zeqi Zheng, Weilong Feng, Dingwei Li, Yingjie Tang, Saisai Wang, Yaochu Jin, Tao Luo, Weng-Fai Wong, Hong Wang, Bowen Zhu. *Spinal-Inspired Artificial Tactile Interneuron with High-Order Burst Spiking for Intelligent Edge Interfaces*. 2026. ***Co-first author***.
5. ***Zhanglu Yan***, Zhenyu Bai, Weng-Fai Wong. *Reconsidering the Energy Efficiency of Spiking Neural Networks*. arXiv:2409.08290, 2024. ***First author***.
6. ***Zhanglu Yan***, Zhenyu Bai, Tulika Mitra, Weng-Fai Wong. *SparrowSNN: A Hardware/Software Co-Design for Energy Efficient ECG Classification*. arXiv:2406.06543, 2024. ***First author***.

Academic Service

- **Conference Reviewer**: ICLR, ICML, AAAI, ICCAD.
- **Journal Reviewer**: IEEE TPAMI, TNNLS.

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